SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – Short Answer

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Short Answer
Weightage:	30 % of CIA	**Total Marks:**	100 Marks
CO1 Statement:	*Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Section / Batch:	CSE(AI) / 2023	**Date:**	24/07/2026

Code of Conduct
I I. V. Growtham Reddy (192372158) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*
Signature of the Student: I. V. Growtham Reddy

Instructions
- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

100%

Questions

Q1. A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. The planning team wants to know how many full-duplex voice channels this spectrum can support.

Q2. A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. During capacity planning, an engineer needs to know how many channels a single cell will typically be allocated.

Q3. A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.

Q4. During a site audit, engineers note the base station radiates 80 W while a nearby handset transmits just 4 W. They want to express this power gap as a decibel-based power control ratio (10-log10 of the ratio).

Q5. A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.

Q6. An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7-cell pattern. How many channels does a single cell end up with?

Q7. A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they now expect?

Q8. Planners want to double a cell's radius while keeping the same SNR at the new edge, in an environment obeying an inverse-square power law. By what factor should transmit power be scaled up?

Q9. A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?

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ITA0301 - Mobile Computing

SLOT - A

CO1 Assessment Tool - 2 - Short Answer

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CSE(AI) / 2023

1. Total Spectrum = 45 MHz = 45,000 KHz

Full duplex voice channel need 2 Simplex channel

$= 2 \times 25 = 50$ KHZ

Channels = $\frac{\cancel{45,000}^{900}}{\cancel{50}} = 900$.

2. Channels per cell = Total channels / N

= 560 / 7

= 80

3. Total voice channels = Carriers × slots/carrier

= 100×8

= 800

4. Power ratio = 80/4 = 20

Ratio in dB = 10× log (20)

= 13.01 dB

≈ 13 dB.

5. C/I (dB) = Signal (dBm) - Interference (dBm)

= -60 - (-92)

= 32 dB

6. Total channels = 42,000 KHz / 300KHz

= 140.

Channels per cell = 140/7 = 20

7. Received power ∝ 1/d^n.

Doubling distance with n = 4 reduces power by 2^4 = 16x

New power = 64/16 = 4 W

8. With n = 2, doubling radius requires power increase by 2^2 = 4x.

9. Total channels in cluster = cluster size × channels/cell

$= 4 \times 15 = 60.$

10. $V = 100$ Km/h $= 27.78$ m/s. Doppler shift $= V \times f/c$

$V \times f/c = 27.78 \times 900 \times 10^6 / 3 \times 10^8$

≈ 83.3 Hz

≈ 83 Hz.

11. slot duration = Frame time / slots

$= 5\text{ms}/8$

$= 0.625$ ms

12. carriers = Total spectrum / spacing

$= \frac{\cancel{24000}}{\cancel{200}}$ 120

$= 120.$

13. TX power in dBm $= 10\log_{10}(8000\text{ mW})$

≈ 39.03 dBm.

Received power $= 39.03 - 33$

$= 6.03$ dBm

≈ 6 dBm

14. using long distance path loss model referenced to 1m Scaling the distance to 2.5 Km give an approximate free space path loss of ≈ 123 dB

15. Total Capacity = calls/cells × cells

$= 250 \times 6.$

$= 1500.$

16. Active calls supported = arrival rate × avg duration

$= 4000$ calls/sec × 180 sec

$= 720{,}000.$

17. Total bandwidth = channels × bandwidth / channel

$= 150 \times 25$

$= 3750$ KHz

$= 3.75$ MHz

No. of. guard bands between 30 bands = 29.

Wasted bandwidth = 29 × 40 KHz

= 1160 KHz

= 1.16 MHz.

19. Speed = 90 km/h = 25 m/s. Cell diameter = 2 × 1.5 km = 3000 m.

Time per cell = 3000/25

= 120 seconds

Handoffs/hour = 3600/120

= 30.

20. Total users = regional centers × base stations / center × users/base station

= 8 × 40 × 250

= 80,000.

21. Approximate cell area ≈ $R^2 = (0.4 km)^2 = 0.16\ km^2$

No. of. cells = 400 / 0.16

= 2500

22. channels = Total spectrum / channel width

= 6000 / 30

= 200

23. Total calls - minutes = 1200 × 4 × 2.5

= 12,000 min/hr

Traffic in Erlangs = 12,000/60

= 200

24. Users = Total bandwidth / bandwidth per user

= 40,000 KHz / 400

= 100

25. Total Capacity = Users / cell × cells

= 600 × 1200

= 720,000

26. Paging operations = Total cells / cells per operation

= 200/4 = 50.

27. Offered traffic = arrival rate × avg duration

= 40,000 × 3 = 120,000 Erlangs

Excess over capacity = 120,000 - 100,000

= 20,000

Fraction blocked = $\frac{20,000}{120,000} \approx 0.17$.

28. Roaming users = 15% × 600,000

= 90,000.

29. Utilization = Peak traffic / Processing rate

= 9,000 / 12,000

= 0.75

= 75%

30. Total channels = Bandwidth / channel width

= 48,000 / 200

= 240

Channels per cell = 240 / 4 = 60.

31. channels = Total spectrum / channel width

= 30,000 / 200

= 150.

32. successfull handoffs = 99% × 2500

= 2475

33. Tx power in dBm = 10 log10 (12000 mW)

≈ 40.79 dBm

Received power = 40.79 - 45

≈ -4.21 dBm

≈ -4 dBm

34. Total capacity = users / cell × cells

= 900 × 7

= 6300.

35. Slot duration = Frame time / slots

= 4.8 ms / 8

= 0.6 ms

36. $V = 150$ Km/h = 41.67 m/s

Doppler shift = $v \times f / c$ = $41.67 \times 1.8 \times 10^{9} / 3 \times 10^{8}$

= 250 Hz

37. Offered traffic = 70000×4 = 280000 Erlangs

Excess = 280000 − 250000

= 30,000

Fraction Blocked = 30000 / 280000

≈ 0.11.

38. Total users in cluster = users/cell × cells in cluster

= 600×9

= 5400

39. Calls dropped/sec = 0.4 % × 60,000

= 240.

40. Path loss (dB) = Tx power (dBm) − Rx power (dBm)

= 43 − (−87)

= 130 dB.

41. Total Coverage = Coverage/station × no. of station

= 2.5×150

= 375 Km²

42. Total channels = 48,000 / 200

= 240

channels per cell = 240/6

= 40.

43. Paging cycles = Total users / users per cycle

= 150,000 / 6000

= 25.

~~44. Hand offs per call = Total users / users per cycle~~

~~= 150000 / 6000~~

~~= 25.~~

44. Handoffs per call = call duration / handoff interval.

$= 9/3$

$= 3.$

45. SNR (dB) = Signal (dBm) − Noise (dBm)

$= -85 - (-115)$

$= 30\,dB.$

46. Minimum bandwidth = users x rate / user

$= 250 \times 150$ kbps

$= 37{,}500$ kbps

$= 37.5$ Mbps.

47. Total call-min/hr = users x calls/hr x avg duration

$= 1500 \times 4 \times 2.5$

$= 15{,}000$ min

traffic in Erlangs $= 15{,}000/60$

$= 250.$

48. Total bandwidth = channels x bandwidth / channel

$= 250 \times 12$

$= 3000$ KHz

$= 3$ MHz.

49. Total throughput = connections x rate / connection

$= 1500 \times 600$ kbps

$= 900{,}000$ kbps

$= 900$ Mbps.

50. Area per tower $= \pi R^2 = \pi \times 4^2 \approx 50.27\,Km^2$

Towers needed $= 350 / 50.27$

$= 6.96$

$= 7.$